

The effects of L1 priming, type of L2 collocation, its frequency, and individuals' L2 proficiency on L2 collocation processing

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Abstract

Aims: This study investigates how Chinese-English bilinguals process English adjective–noun collocations under different priming conditions to understand the mechanism underlying the L1 congruency effect. By examining how various factors affect processing and knowledge representation, this research aims to shed light on effective language teaching strategies.

Methodology: We employed cross-language masked priming followed by an acceptability judgment task with two groups of participants: lower-intermediate (94) and advanced (74) Chinese-English learners.

Findings: (1) The statistics showed that collocation type-specific response times for these groups differed; only the advanced group critically manifested the L1 Chinese congruency effect in response time under the Chinese-congruent priming condition. (2) Lower-intermediate learners had limited knowledge representation of all three types (“Chinese-congruent,” “English-only,” “Chinese-only”), while advanced learners developed better knowledge of “Chinese-congruent” and “English-only” collocations, but not “Chinese-only” ones. (3) Furthermore, while lower-intermediate learners showed sensitivity only to the frequencies of “Chinese-congruent” ones, the advanced learners became sensitive to the frequencies of not only “Chinese-congruent” but also “English-only” ones.

Originality: This study contributes to our understanding of the collocation processing mechanism by comparing results from a wider range of proficiency levels (lower-intermediate vs. advanced) rather than focusing narrowly on intermediate versus advanced learners. The study’s methodology combines cross-language masked priming with an acceptability judgment task, allowing for a more nuanced understanding of how bilinguals process different types of adjective–noun collocations.

Implications: This research highlights the importance of L2 exposure in shaping bilinguals’ mental representation of English adjective–noun collocations and in forming L1 congruency effect.

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The findings have implications for improving L2 collocation instruction strategies by relying less on supposedly vital L1 influence but more on exposure to any individual collocation to learn.

Keywords

L2 adjective–noun collocation, L1 semantic activation, cross-language masked priming, L1 congruency effect

Introduction

Second language (L2) learners often struggle with processing collocations that are specific to a target language (González Fernández & Schmitt, 2015; Henriksen, 2013; Sonbul & El-Dakhs, 2020). To avoid terminology confusion, we define collocations as two-word combinations that are grammatically well-formed and frequently co-occurring pairs with strong association strength (Mutual Information [MI] > 3) as measured by native-speaker corpus-based frequency analysis following Kjellmer (1994) and Stubbs (1995). They are distinct from fixed idioms (e.g., *kick the bucket*), chunks (e.g., *in front of*), and word strings (*in the middle of the*), though some researchers used them interchangeably (e.g., Wray, 2002). Collocations are governed by convention, making the association between the constituent words largely arbitrary; changing one word can significantly alter its meaning or reference (Wolter & Yamashita, 2015; Yamashita, 2018). For instance, the adjective *major* in the collocation *major earthquake* cannot be replaced by *big* to convey exactly the same collocation meaning of *major*.

We also recognize that across languages some collocations may be “L1-congruent” (i.e., having word-for-word translation equivalents in L1 such as *big eyes* having exact literal translation 大眼睛 in L1 Chinese), while others are “X-language-only,” where literal translation of a component word does not yield its equivalent meaning (Bahns, 1993; Wolter & Gyllstad, 2013). For instance, *hot pepper* cannot be literally translated into 热椒 in L1 Chinese. A third type of collocations is pseudo “L1-only” collocations that researchers have invented to serve as baseline items for effective comparison of processing effects. For instance, **hot heart* does not exist in English, but Chinese-English learners may translate 热心 that exist in Chinese into **hot heart* word for word when they actually mean *warm-hearted*.

Research into the processing of “L1-congruent” collocations, “L2-only” collocations, and non-collocations (e.g., **narrow ability*) provides an opportunity to explore the development of L2 learners’ mental representation (knowledge) of the different types of collocations (Sonbul & El-Dakhs, 2020; Wei, 2003; Yamashita & Jiang, 2010).

Research has shown that first language (L1) semantic activation can influence the processing of these linguistic units in L2 contexts, leading to the so-called L1 congruency effect (processing advantage of “L1-congruent” collocations over “L2-only” collocations) in acceptability judgment or lexical decision tasks (e.g., Wolter & Gyllstad, 2011; Zhang, 2017). Frequency effect has also been observed, with the more frequently encountered collocations being processed faster and more accurately than the less common ones (e.g., Wolter & Gyllstad, 2013). In addition, estimated proficiency has also been found to modulate the L1 congruency effect (Sonbul & El-Dakhs, 2020).

Which is more responsible for L1 congruency effect is a question not yet well-investigated so far. In this study, we are interested in how L2 English learners at two proficiency levels process different types of English adjective–noun collocations along a frequency continuum. Specifically, we aim to explore whether and how these types of collocations differ in processing under different cross-language L1 Chinese priming conditions.

L1 congruency effect

Despite previous research indicating that learners exhibit an advantage in processing “L1-congruent” collocations, which is often referred to as the L1 congruency effect (e.g., Wolter & Gyllstad, 2011; Wolter & Yamashita, 2018; Zhang et al., 2017), there wasn’t always a significant advantage for “L1-congruent” collocations over “L2-only” collocations. For instance, in acceptability judgment tasks, Chen (2023) and Yamashita and Jiang (2010) observed a congruency effect in both processing accuracy and speed with non-immersive English learners, but only an accuracy congruency effect with immersive English learners; in contrast, Fang and Zhang (2021) found a congruency effect in processing accuracy but not in speed in a collocation acceptability judgment task with both university undergraduate (intermediate) and graduate (advanced) non-immersive English learners. Some research has suggested that the processing advantage associated with L1 congruency may be more nuanced than previously thought and that collocation properties (e.g., frequency and cross-language similarity/congruency) and individual differences in learner characteristics (e.g., L2 proficiency) can influence the strength of this effect (e.g., Chen, 2023; Fang and Zhang, 2021; Sonbul & El-Dakhas, 2020; Yamashita and Jiang, 2010; Zhang & Fang, 2020). This ambiguity highlights the need for further investigation into the L1 congruency effect to fully understand the underlying mechanism.

Frequency effect

Despite the recognized importance of frequency in language learning and teaching, surprisingly little research has explored how learners respond to collocations along the frequency continuum. Research suggests that high-frequency collocations, which are processed faster, are more likely to be mentally represented in learners’ lexicon than low-frequency ones (Wolter & Gyllstad, 2013; Wolter & Yamashita, 2018).

However, previous studies on collocation frequency have primarily focused on the competition between collocation frequencies and constituent word frequencies among more proficient learners to investigate whether collocations are processed holistically as one single unit or analytically (Fang & Zhang, 2021; Lou, 2022; Öksüz et al., 2021; Sonbul, 2015). These investigations have shed light on how learners with higher proficiency levels process collocations of varying frequencies. Nevertheless, there is a notable gap in our understanding of how less proficient L2 speakers respond to such a range of collocations.

In particular, few studies have begun to investigate whether low- and intermediate-level L2 learners mentally represent high-frequency collocations (Wang, 2015). However, Wang’s study has employed categorical frequency bands, which can lead to inaccurate modeling and prediction in statistical analysis due to the discrete nature of frequency categories. Moreover, there is a scarcity of collocation frequency research with such learners, leaving us uncertain about whether frequencies based on native-speaker corpora are relevant to predicting how low- and intermediate-level L2 learners process collocations (Hong & Ma, 2023).

Furthermore, the existing body of knowledge has not fully explored how L2 learners with wider-ranging language proficiency levels respond differentially to collocation frequencies across various types of collocations. This lack of understanding creates a clear opportunity for research into whether learner sensitivity to collocation frequencies is collocation type-specific.

Proficiency effect

The relationship between L1 congruency and L2 proficiency has been explored in various studies (Chen, 2023; Fang & Zhang, 2021; Sonbul & El-Dakhas, 2021; Wolter & Yamashita, 2018; Yamashita & Jiang, 2010; Zhang & Fang, 2020), but findings on this topic did not converge. For

instance, while Wolter and Yamashita (2018) suggested that L1 congruency affects response times among English learners at both intermediate and advanced levels of proficiency, Fang and Zhang (2021) found no significant L1 congruency effect in response time at intermediate and advanced levels. A key limitation in these two studies is the narrow range of proficiency levels examined, with all the participants being university-level students. In contrast, some research has explored how intermediate in a non-immersive learning setting and advanced English learners in an immersive learning setting would process collocations (congruent vs. incongruent) (Chen, 2023; Lee, 2021), but the learning environment difference may have introduced an extra confounding factor.

These limitations highlight the need for further research on low- and intermediate-level learners. Low- and intermediate-level learners in the non-immersive contexts were particularly understudied. While Lee (2021) targeted learners at a wider range of proficiency levels (lower-intermediate vs. advanced) in the inquiry into the relationship between proficiency and L2 collocation recognition, there is still a need for further research to include L2 learners at such earlier stages of development. This makes it possible to investigate “whether proficiency affects the degree to which L1 influence occurs in the recognition of L2 collocations,” as Lee (2021: 7) noted. In a word, further research on lower- and intermediate-level learners could provide valuable insights into how L1 congruency influences L2 collocation recognition.

Theoretic approach to L1 congruency effect

While the L1 semantic activation theory (Jiang, 2000a) remains one of the most influential explanations for the L1 congruency effect, research suggests that it may not fully account for all findings in previous studies. According to this theory, learners rely on L1 mediation when learning new L2 vocabulary items, which can lead to an automatic processing advantage for “L1-congruent” collocations due to the strong association between words sharing similar semantic and syntactic features across L1 and L2. This theory also predicts that components of “L1-congruent” collocations will be more strongly associated than those of “L2-only” collocations at the initial learning stage, and that there will be a similar advantage for “L1-only” collocations over “L2-only” collocations.

However, this prediction is not consistently supported by empirical evidence. While some studies (e.g., Zhang et al., 2017) have found that both “L1-congruent” and “L1-only” collocations exhibit processing advantages over “L2-only” collocations, others (Lee, 2021; Ren & Wang, 2020; Wolter & Yamashita, 2015, 2018; Zhan & Ni, 2024) have reported a partially different pattern of results: the congruency effect with the “L1-congruent” collocations is maintained, but “L1-only” collocations do not have a processing advantage over “L2-only” collocations. This discrepancy highlights a need for further research to verify whether “L1-only” collocations as well as “L1-congruent” collocations indeed exhibit processing advantages over “L2-only” collocations due to L1 influence.

Besides, in an attempt to better mirror the complex mechanism underlying L1 congruency effect, Yamashita and Jiang (2010), for instance, extend the theory to include an additional factor influencing learners’ processing of collocations: frequency of exposure to L2 collocations as proficiency increases. According to this perspective, actual exposure frequency becomes a dominant determinant in L2 collocation processing at higher levels of proficiency, leading the L1 congruency effect to gradually diminish. Whether the L1 congruency effect is strongest, as supposed, at an earlier stage (lower-intermediate) remains an empirical question that warrants further investigation.

Justification for methodological adjustment to masked primed acceptability judgment task

To investigate the L1 congruency effect, online behavioral experimental paradigms have been widely employed. For instance, lexical decision tasks (e.g., Wolter & Gyllstad, 2011; Wolter &

Yamashita, 2015; Zhang, 2017) allow researchers to examine how quickly and accurately participants can identify collocations as whole lexical units, while acceptability judgment tasks (e.g., Gyllstad & Wolter, 2016; Wolter & Gyllstad, 2013; Yamashita & Jiang, 2010) measure the accuracy and speed of recognizing meanings in collocations. However, these paradigms may not be sufficiently sensitive to capture more nuanced changes in semantic representation involved in processing collocations.

Our pilot studies using a lexical decision task and an acceptability judgment task yielded inconsistent results regarding L1 congruency effects. Just as Wolter and Yamashita (2015) did not reveal expected L1 activation for “L1-only” collocations, we also failed to find processing advantages for “L1 Chinese-only” collocations. Considering the inconsistent findings from the lexical decision and acceptability judgment tasks and the suggestion from Sonbul & El-Dakhs (2020) of employing alternative measures beyond the two above-mentioned tasks, previous research highlights the need for methodological adjustments.

Preliminary research evidence supports the adoption of an adjusted cross-language primed acceptability judgment task to better measure L1 congruency effects. Cross-language masked priming involves a specific type of semantic repetition priming between translation equivalents, where word pairs (e.g., 狗-*dog*) are presented sequentially in tasks such as lexical decision or word naming on the second word “*dog*” (the equivalent of 狗 in Chinese) compared to semantically unrelated cross-language word pairs (e.g., 箱子 [Chinese for *box*]-*dog*). The reliable semantic repetition effect from these translation equivalents signals shared conceptual representation across languages (Zeelenberg & Pecher, 2003).

The priming effect of L1 translation equivalents on L2 words is particularly robust among both early and late proficient bilinguals (de Groot, 1992; Gollan et al., 1997; Jiang, 2000b; Perea et al., 2008). Moreover, the literal-translation priming effect between 狗 and *dog* is stronger than that between semantically related pairs across languages (e.g., 狗-*cat*) (Zhao & Li, 2013), highlighting robust conceptual representation of individual L2 words in learners’ mental lexicon.

Cao (2018) pioneered the use of cross-language priming combined with acceptability judgment to investigate the mental representation of L2 collocations. He examined how Chinese L1 stimulus primes—correct L1 free translations for L2 target collocations (e.g., 水平 [*level*]), semantically related terms (e.g., 成本 [*cost*]), and semantically unrelated terms (e.g., 窗口 [*window*])—affect the processing of “L2 English-only” collocations like “*living standard*.” The free-translation priming effectively teased out mental representation of these “English-only” collocations. Methodologically, participants saw a prime (e.g., 水平) for 300 ms before making acceptability judgments on the target collocation. This psycho-linguistic priming paradigm enabled the comparison of processing effects on the same target collocation under the three priming conditions. Such a prime-target design can effectively control for potential confound arising from processing discrete target collocations according to Jiang (2012). In addition, Cao’s study also illustrates the effective use of L1 free-translation equivalents of L2-only collocations to obtain a cross-language priming effect. L2-only collocations cannot be literally translated without distorting collocational meaning. The semantically equivalent L1 free translation also helps determine whether the mental representation of L2-only collocations has been established.

However, Cao’s study used a prime-target interval of 300 ms, which may not lead to automatic semantic processing. Wang and Zhang (2013) recommend using masking techniques and shortening the presentation time for primes to within 60 ms, to ensure that participants are unaware of them.

Therefore, the cross-language masked priming paradigm followed by an acceptability judgment should be conducive to tapping into how L2 collocations are processed and conceptually represented.

To sum up the literature, research has shown that L1 congruency, L2 collocation frequency, and L2 proficiency all play a role in collocation processing. However, more research is needed to examine the complex interactions among the multiple factors. Attempts to update the L1 activation account of the congruency effect underscore the importance of considering both L2 proficiency (learner individual factor) and collocation frequency (corpus distribution) that influence L2 learners' collocation processing. Understanding the mechanism underlying the L1 congruency effect in collocation processing has implications for L2 collocation learning and instruction. This research will provide new insights into the complex interactions among the different factors. To better understand this complex phenomenon, we adopt a modified research paradigm that combines cross-language masked priming with an acceptability judgment task and aim to further explore whether and how the three factors including L1 Chinese semantic priming type, L2 English collocation frequency, and L2 English proficiency affect the processing of the three types of "Chinese-congruent," "Chinese-only," and "English-only" adjective–noun collocations. We choose to focus on the processing of adjective–noun collocations so as to do cross-study comparison (e.g., Chen, 2023; Sonbul & El-Dakhas, 2021; Wolter & Gyllstad, 2013; Wolter & Yamashita, 2018; Yamashita & Jiang, 2010; Zhang & Fang, 2020).

In such a modified research paradigm, we may better address L2 collocation processing by examining the effects of L1 translation priming, the type of L2 collocation, its frequency, and individuals' L2 English proficiency in one research setting, which has not been so approached in previous research on L2 collocation processing.

The current study

Research questions

Based on the literature review above, we addressed the following questions in an experiment task combining cross-language masked priming and semantic acceptability judgment:

RQ1. Do lower-intermediate and advanced English learners both accept "Chinese-congruent" and "Chinese-only" adjective–noun collocations at higher rates and faster than "English-only" ones? Does this congruency effect (if any) decrease as learner proficiency increases?

RQ2. Do English learners at lower-intermediate and advanced levels both exhibit mental representation of "Chinese-congruent," "Chinese-only", and "English-only" adjective–noun collocations?

RQ3. Can native-speaker corpus frequency predict how quickly lower-intermediate and advanced English learners will respond when making acceptability judgments about adjective–noun collocations?

More of the previous research in this field suggests that there should be L1 congruency effect, where "L1-congruent" collocations are processed with an advantage over "L2-only" ones at higher proficiency levels—specifically among advanced learners in non-immersive foreign language learning settings. Based on this, we hypothesized that at the advanced level, English learners would accept "L1 Chinese-congruent" collocations at a higher rate and/or faster than "L2 English-only" ones. In line with Jiang's (2000a) L1 semantic activation theory, which posits an even stronger L1 congruency effect for learners at earlier stages of L2 learning, we also anticipated that lower-intermediate learners would exhibit an even stronger L1 congruency effect compared to advanced university students. Specifically, we expected that these learners would show a processing advantage of "L1 Chinese-congruent" and "L1 Chinese-only" collocations over "L2 English-only" ones,

although a handful of studies suggest that “L1-only” collocations may not have a processing advantage over “L2-only” ones with the lower-level (intermediate university students) group.

In light of the dominant L1 semantic activation theory and empirical findings concerning L2 collocation processing, we assumed that there would be an automatic transfer of L1 Chinese collocation knowledge into L2 English collocations. Specifically, we predicted that English learners would have intrinsic knowledge and mental representation of both “Chinese-congruent” and “Chinese-only” collocations but not for “English-only” ones. As learner proficiency increases, however, their knowledge of “English-only” collocations would also grow alongside that of the other two types of collocations.

Since the extended L1 semantic activation theory (Yamashita & Jiang, 2010) incorporates exposure frequency of L2 collocations as a key factor affecting the L1 congruency effect, we expected advanced learners to exhibit both the L1 congruency effect and corpus frequency sensitivity for “Chinese-congruent” and “English-only” collocations. However, given that native-speaker corpus frequency sensitivity typically occurs only after a certain threshold of learner exposure (Wolter & Gyllstad, 2013), we expected no significant variation in response times for “Chinese-only” collocations across both levels due to their extremely low encounter rate in L2 learning contexts.

Experimental materials

Material collection. We first collected 330 high-frequency Chinese monosyllabic adjectives from the BCC (Beijing Language and Culture University’s Chinese Corpus). To identify adjective words that have all the three types of “Chinese-only,” “Chinese-congruent,” and “English-only” adjective–noun collocations, we then consulted Chinese dictionaries (*the Modern Chinese Dictionary*, *the Chinese 8000 Words Dictionary*, (Lexicography Office of CASS Language Institute, 2016) and the Vocabulary Syllabus of HSK [the Chinese Proficiency Test]) (BLCU Chinese Proficiency Test Center, 2000) and English dictionaries (*the online Collins*, *Oxford*, and *Free dictionaries*). By comparing dictionary entries across these resources, we eliminated adjectives without all the three types of adjective–noun collocations. As a result, we selected 30 adjectives with “Chinese-only” (e.g., *hot heart**), “Chinese-congruent” (e.g., *hot summer*), and “English-only” (e.g., *hot pepper*) collocations for further filtering.

We further checked the corpus frequencies of individual adjective words in both Chinese (BCC) and American English (COCA) corpora. To ensure no significant word frequency difference within each language, we log transformed the raw word frequencies of the adjectives from both corpora following Wolter and Gyllstad’s (2013) methodology. For 20 of the adjectives, there was no significant log-transformed word frequency difference within each language: $M_{\text{Chinese}} = 5.667$, $t = -0.254$, $p = .802$; $M_{\text{English}} = 4.932$, $t = 0.006$, $p = .996$.

Finally, we inspected the collocation frequency and MI score of each adjective–noun collocation in COCA and calculated the t -score for each collocation. The final nine pairs of “Chinese-congruent” and “English-only” collocations showed no significant differences in frequency ($t = -0.076$, $p = .941$), with appropriate t -scores (>2) and MI scores (>3), based on Stubbs’ (1995) criteria. The corresponding pairs had t -scores (4.13–55.91, $t = 0.652$, $p = .533$) and MI scores (3.58–6.42, $t = 0.171$, $p = .869$). The Chinese-only collocations were minimally found to have rare occurrences in COCA. For the final collocations selected, refer to the Supplementary Materials document.

Participants

In this study, we recruited two groups of participants: 94 high-school second-year students and 74 English-graduate students who had passed TEM-Band 8 (TEM for Test for English Majors designed specifically for university students majoring in English across China). According to their

alignment with the Chinese Standards of English Language Ability (CSE) and the Common European Framework of Reference for Languages (CEFR), as per Liu and Yang (2021), high-school students are typically at a lower-intermediate level, while TEM-Band 8 passers fall at an advanced level. Therefore, these two groups were categorized as lower-intermediate and advanced.

To estimate their English proficiency on one scale, we followed Sonbul and El-Dakhs (2020), Wolter and Gyllstad (2013), and Wolter and Yamashita (2018), using a single vocabulary test across both groups since vocabulary size scores are associated with L2 proficiency (Milton, 2010).

The Vocabulary Size Test (VST) by Nation and Beglar (2007) revealed significant differences in estimated proficiency between the lower-intermediate group ($M_{age} = 16.330$ years; $SD = 0.594$, $M_{vocabulary} = 47.957$; $SD = 3.787$) and the advanced group ($M_{age} = 23.934$ years; $SD = 1.883$, $M_{vocabulary} = 84.905$; $SD = 10.079$). The F -statistic for the estimated proficiency (VST range = 44–62 for lower-intermediate, 64–111 for advanced) was 1,072.316 ($p < .0001$), with an effect size of $\eta^2 = 0.866$.

Participants in both groups studied English primarily in formal classroom settings within China and had no overseas experience, mainly differing by estimated proficiency level: lower-intermediate or advanced.

Experimental paradigm and procedure

The experiment task included cross-language masked L1 priming followed by collocation acceptability judgment. The mixed design had four factors of theoretical interest: English proficiency (two levels, between-group) * priming condition (three levels, within-group) * target collocation type (three levels, within-group) * collocation frequencies (continuous). To ensure that each participant saw an equal number of prime-target collocation pairs across three different priming conditions, the prime-target collocation pairs were equally divided into three counter-balanced lists (List A, List B, List C) according to the Latin square design. Each list consisted of 27 prime-target collocation pairs (3 prime conditions * 9 collocations) and 135 prime-filler collocation pairs (McNamara, 2005; Neely et al., 1989).

The three language lists were randomly assigned within each group. For the intermediate group, List A was allocated to approximately one-third (33 participants), List B to another third (30 participants), and List C to the remaining third (31 participants). Similarly, for the advanced group, List A was given to 25 participants, List B to another 25, and List C to the last 24. In addition, the item trial order in the experiment was randomized.

The presentation sequence began with a fixation point “+” lasting for 1,500 ms, followed by a masking symbol ##### that lasted for 500 ms and then the prime word appeared for 60 ms before a blank screen was displayed for 100 ms. The target English collocation then appeared at the center of the screen, prompting participants to make a binary judgment on its acceptability within 4,000 ms—either by pressing “J” for acceptance or “F” for rejection. If no response was made within this time limit, it would time out and get logged as “non-response” in E-Prime 2.0.

Table 1 showcases one set of nine prime-target collocation combinations using *hot* collocations as an example. Following Fang and Zhang (2021), the target collocations in each set share the same node (“*hot*”) to minimize confounding influences from item heterogeneity, and their accompanying collocates do not have ambiguous meanings. The collocations (each consisting of the same node and a different collocate) fall into three types—“Chinese-congruent,” “English-only,” and “Chinese-only.”

The L1 Chinese primes were designed such that they either shared the same semantic concept with or did not semantically relate to the target collocations:

Table 1. Priming conditions for the set of *hot* collocations.

Priming conditions in Chinese			
Chinese-congruent	English-only	Semantically unrelated	Type of target English collocation
热 (rè; hot)	辣 (là; spicy)	甲 (jiǎ; first in order)	<i>hot summer</i> (“Chinese-congruent”)
热 (rè; hot)	辣 (là; spicy)	甲 (jiǎ; first in order)	<i>hot pepper</i> (“English-only”)
热 (rè; hot)	辣 (là; spicy)	甲 (jiǎ; first in order)	<i>*hot heart</i> (“Chinese-only”)

Chinese-congruent primes: “Chinese-congruent” and “Chinese-only” collocations share the prime word labeled as the Chinese-congruent prime, which is their common literal translation in Chinese for the node *hot* (e.g., 热 *rè* means *hot in temperature* in the “Chinese-congruent” collocation *hot summer*, and *willing to help* when paired with 心 *xīn*; heart) in the “Chinese-only” collocation **hot heart*.

English-only primes: an “English-only” collocation (e.g., *hot pepper*) has only a free-translation equivalent in Chinese (e.g., 辣-*là*; spicy) to match the semantic concept for the node in the collocation. To distinguish from the above Chinese-congruent prime word type, the primes using free translations of “English-only” collocations were named English-only primes.

Semantically unrelated primes: These L1 Chinese primes semantically unrelated to the target collocations served as baseline priming conditions. These primes used common numerical words from ancient Chinese astronomical calendars (e.g., 甲 *jiǎ* first in order), which have restricted meanings and form a homogeneous set of semantically unrelated prime words.

As shown in Table 1, each collocation was primed under all three conditions: Chinese-congruent, English-only, and semantically unrelated. Since the target collocations within the same set share the same node (e.g., *hot* for *hot summer*, *hot pepper*, **hot heart*), this allows us to compare how participants process different collocations under identical priming conditions. When prime and target match, a strong translation priming effect for an existing L2 collocation is expected due to shared semantic concepts across languages. What’s more, by observing participant processing of the same collocation under these three priming conditions, we can infer how each collocation is represented in their mental lexicon (refer back to section “Justification for methodological adjustment to masked primed acceptability judgment task”).

Learner bio-data collection and vocabulary size testing were arranged before the main experiment. Concerns about missing data in the main experiment due to timeouts (non-responses) were addressed through the use of mixed modeling in R, which allowed for a more accurate analysis of participants’ response times and binary acceptability judgments. E-Prime 2.0 automatically logged these data points after each trial.

Data analysis

Data analysis was based on response times and binary (1, 0) data from the acceptability judgment task. “Accepted” responses were assigned a score of one, while “rejected” responses were assigned a score of zero, following Wolter and Yamashita (2015). Collocations with zero raw corpus frequency in the “Chinese-only” type were uniformly converted to one for log transformation as described by Wolter and Yamashita (2018).

The *lmer* and *glmer* functions from the *lme4* package of R software (R Core Team, 2022) were used to model response time and binary acceptability judgment data separately for each

proficiency group according to Bates et al. (2015). In the initial models, the following variables and their interactions were included as fixed-effect factors: priming type, collocation type, log-transformed vocabulary size scores, collocation frequency, collocation length, and trial order.

Random-effect factors consisted of participants and test items. A maximal random-effects structure model was built and simplified until it best fit the data following the principle of “*keep it maximal*” (Barr et al., 2013; Ma et al., 2022). Initially, all random intercepts and random slopes for subjects (participants) and items (collocations) were included, and then the random structure was stepwise simplified using maximum likelihood estimation.

The *vif.mer* function was used to check for multicollinearity in the model, ensuring that variance inflation factor (VIF) coefficients were within a reasonable range as per Hair et al. (1995) and Kapteijns and Hintz (2021).

The *scale* function from R software was applied to mean center and standardize the log-transformed continuous variables (vocabulary size scores, collocation frequency, collocation length, and trial order). Fixed-effect factors’ main effects and interaction effects were obtained using the *mixed* function from the *afex* package. Post hoc tests were conducted with *emmeans* and *emtrends* functions.

Results

Descriptive statistics

Table 2 presents the descriptive statistics of the response times and acceptance rates by type of English adjective–noun collocations in each of the two proficiency groups. The most notable feature is that the acceptance rates decreased from “Chinese-congruent,” to “English-only” to “Chinese-only” collocations almost in all the priming conditions, regardless of proficiency.

For the lower-intermediate group, there was little variation in the response times of the three types of collocations across the priming conditions, but for the advanced group, the average response time of the “Chinese-only” collocation was numerically longer than the other two types under all priming conditions. In addition, within the advanced group, response times for “Chinese-congruent” and “English-only” collocations were shortest in their respective priming conditions. With both groups, the response times and acceptance rates varied by priming condition.

Model analyses

Given a collinearity issue (with VIF values exceeding 7.9) in our large model when analyzing data from both intermediate and advanced participants, we developed separate regression models for each group.

Model analyses of the acceptability judgment with the lower-intermediate group revealed that the best mixed-effects model had VIF values under 4.85, and there was a significant main effect for priming condition, $\chi^2(2)=8.62, p=.013$, and an interaction effect for priming condition*collocation type, $\chi^2(4)=13.3, p=.01$; the best mixed-effects model on response time showed VIF values under 5.6, and there was a significant main effect for collocation frequency, $\chi^2(1)=4.66, p=.031$, and an interaction effect for collocation type*collocation frequency, $\chi^2(2)=7.68, p=.022$.

With the advanced group, the best mixed-effects model of acceptance judgment had VIF values under 4.46, and there were significant effects of collocation frequency, $\chi^2(1)=4.95, p=.026$, and an interaction effect for the priming condition*collocation type, $\chi^2(4)=17.9, p=.001$; the best response time model had VIF values under 5.298, and there were significant main effects of priming condition, $\chi^2(2)=7.61, p=.022$, and collocation frequency, $\chi^2(1)=11.68, p<.001$, and

Table 2. Descriptive statistics of participants' response times and acceptance rates in the judgment task.

English proficiency	Priming conditions	"Chinese-congruent" collocations		"English-only" collocations		"Chinese-only" collocations	
		RT (SD)	Acceptance rate (SD)	RT (SD)	Acceptance rate (SD)	RT (SD)	Acceptance rate (SD)
Lower-intermediate	Chinese-congruent	7.07 (0.52)	74.8 (43.50)	7.19 (0.58)	56.5 (49.7)	7.15 (0.59)	54.0 (49.9)
	English-only	7.11 (0.51)	71.2 (45.40)	7.20 (0.51)	69.4 (46.2)	7.18 (0.54)	56.3 (49.7)
	Semantically unrelated	7.13 (0.52)	69.9 (45.90)	7.18 (0.53)	55.5 (49.8)	7.13 (0.58)	50.7 (50.1)
Advanced	Chinese-congruent	7.1 (0.39)	76.5 (42.50)	7.27 (0.38)	57.7 (49.5)	7.33 (0.44)	30.9 (46.3)
	English-only	7.3 (0.38)	64.7 (47.90)	7.14 (0.40)	65.1 (47.8)	7.36 (0.44)	33.9 (47.5)
	Semantically unrelated	7.23 (0.39)	73.7 (44.10)	7.24 (0.39)	56.2 (49.7)	7.43 (0.42)	29.6 (45.8)

Note. RTs (response times) are log transformed. Acceptance rates are presented as percentages (%). Since "Chinese-only" collocations are non-collocations in English, we use the term "acceptance" instead of "accuracy" in the subjective judgment task.

interaction effects of collocation type*collocation frequency, $\chi^2(2)=10.42$, $p=.005$, and priming condition*collocation type, $\chi^2(4)=36.17$, $p<.001$.

In summary, the acceptance rate decreased from "Chinese-congruent" to "English-only" to "Chinese-only" collocations across both proficiency groups. For the advanced group, response times varied significantly both by priming condition and by collocation frequency; interaction effects between priming condition and collocation type, as well as collocation type and collocation frequency, were also significant.

Results addressing RQ1. For RQ1, do lower-intermediate and advanced English learners both accept "Chinese-congruent" and "Chinese-only" adjective–noun collocations at higher rates and/or faster than "English-only" ones? Does this congruency effect (if any) decrease as learner proficiency increases?

It was anticipated that at the advanced level, English learners would accept more "Chinese-congruent" collocations or faster than "English-only" ones. For lower-intermediate learners, we expected an even stronger L1 congruency effect (i.e., a processing advantage of "Chinese-congruent" collocations) compared to advanced university students and a possible advantage for "Chinese-only" collocations over "English-only" ones.

The results from Table 3 show that the lower-intermediate group accepted significantly fewer "English-only" collocations than "Chinese-congruent" collocations under both the unrelated priming condition ($p=.043^*$) and the Chinese-congruent priming condition ($p=.009^{**}$). However, they did not react any faster to any type of collocation under any priming condition.

In contrast, advanced learners exhibited a response time congruency effect between "Chinese-congruent" versus "English-only" collocations only under the Chinese-congruent priming condition ($p=.017^*$). They did not show this congruency effect for "Chinese-only" versus "English-only" collocations.

In addition, there was only a close-to-significant processing advantage for "English-only" collocations over "Chinese-congruent" ones under the English-only priming condition ($p=.066$).

Overall, L1 congruency effects were only observed in terms of response time with advanced learners under Chinese-congruent priming condition. Lower-intermediate learners didn't exhibit a stronger L1 congruency effect than advanced learners, which contradicts expectations based on the

Table 3. The results of post hoc analyses for processing different types of collocations in the same priming condition in the respective models by participant group.

Level	Priming condition	Collocation-type contrast	Acceptance rate			Response time		
			Estimate	z	p	Estimate	t	p
Lower-intermediate	Unrelated	“Chinese-congruent” vs. “Chinese-only”	0.68	0.99	.325	-0.07	-0.78	.443
	Unrelated	“Chinese-congruent” vs. “English-only”	0.85	2.02	.043*	-0.02	-0.40	.697
	Unrelated	“Chinese-only” vs. “English-only”	0.17	0.24	.812	0.05	0.52	.610
	Chinese congruent	“Chinese-congruent” vs. “Chinese-only”	0.83	1.21	.227	-0.07	-0.78	.443
	Chinese congruent	“Chinese-congruent” vs. “English-only”	1.09	2.61	.009**	-0.02	-0.40	.697
	Chinese congruent	“Chinese-only” vs. “English-only”	0.26	0.37	.712	0.05	0.52	.610
	English-only	“Chinese-congruent” vs. “Chinese-only”	0.38	0.55	.585	0.07	-0.78	.443
	English-only	“Chinese-congruent” vs. “English-only”	0.07	0.17	.866	-0.02	-0.40	.697
	English-only	“Chinese-only” vs. “English-only”	-0.31	-0.43	.665	0.05	0.52	.610
	English-only	“Chinese-only” vs. “English-only”	-0.31	-0.43	.665	0.05	0.52	.610
Advanced	Unrelated	“Chinese-congruent” vs. “Chinese-only”	0.47	0.49	.626	-0.10	-0.71	.482
	Unrelated	“Chinese-congruent” vs. “English-only”	0.98	1.70	.090	-0.03	-0.43	.671
	Unrelated	“Chinese-only” vs. “English-only”	0.52	0.52	.602	0.06	0.48	.634
	Chinese congruent	“Chinese-congruent” vs. “Chinese-only”	0.51	0.53	.599	-0.15	-1.11	.277
	Chinese congruent	“Chinese-congruent” vs. “English-only”	1.00	1.73	.083	-0.18	-2.55	.017*
	Chinese congruent	“Chinese-only” vs. “English-only”	0.50	0.51	.614	-0.04	-0.27	.793
	English-only	“Chinese-congruent” vs. “Chinese-only”	-0.36	-0.37	.709	0.04	0.31	.757
	English-only	“Chinese-congruent” vs. “English-only”	-0.21	-0.37	.710	0.14	1.91	.066
	English-only	“Chinese-only” vs. “English-only”	0.15	0.15	.883	0.10	0.72	.478
	English-only	“Chinese-only” vs. “English-only”	0.15	0.15	.883	0.10	0.72	.478

Note. Significant differences are indicated with the convention of * $p < .05$, ** $p < .01$, *** $p < .001$.

L1 semantic activation theory. Moreover, there was no evidence of processing advantages for “Chinese-only” collocations over “English-only” ones at either proficiency level.

Results addressing RQ2. For RQ2, do English learners at lower-intermediate and advanced levels exhibit mental representation of “Chinese-congruent,” “Chinese-only,” and “English-only” adjective–noun collocations?

We predicted that English learners would have mental representations of both “Chinese-congruent” and “Chinese-only” collocations from the very beginning, while initial knowledge of “English-only” collocations might be limited or non-existent. As learner proficiency increases, we expected a growth in knowledge for all types of collocations, with translation priming being stronger than priming under the baseline semantically unrelated condition if mental representations were present:

Lower-intermediate group: It was observed (see Table 4) that the lower-intermediate group showed no significant differences between acceptance rates for “Chinese-only” collocations under both unrelated and Chinese-congruent priming conditions. Similarly, there were no differences in acceptance rates for “Chinese-congruent” collocations across these two conditions. However, “English-only” collocations demonstrated significantly lower acceptance rates under the semantically unrelated condition compared to English-only priming condition ($p = .0002^{***}$). Nonetheless, “English-only” collocations were not more accepted than “Chinese-only” and “Chinese-congruent” collocations in any condition (see Figure 1). For the collocations accepted, response times by the lower-intermediate learners for any type of collocations did not differ between translation priming and semantically unrelated conditions. This suggests minimal mental representation of all the types of collocations at this stage (see Figure 2).

Advanced group: At this stage, neither acceptance rates nor response times for “Chinese-only” collocations changed significantly between the unrelated versus Chinese-congruent priming condition. However, significant differences were observed in the acceptance rate difference for “English-only” collocations between English-only and semantically unrelated conditions ($p = .014^*$).

What’s more, processing of “Chinese-congruent” collocations was significantly faster under the Chinese-congruent priming condition compared to the semantically unrelated one ($p = .0001^{***}$). Similarly, “English-only” collocations were processed more efficiently in the English-only priming condition than in the baseline unrelated condition ($p = .013^*$), indicating strong mental representations of both types at this stage.

These findings suggest that advanced learners have robust lexical representations for “Chinese-congruent” and “English-only” collocations, whereas lower-intermediate learners lack these representations or exhibit minimal knowledge of all the three types of collocations.

Results addressing RQ3. For RQ3, can native-speaker corpus frequency predict how quickly lower-intermediate and advanced English learners will respond when making acceptability judgments about adjective–noun collocations?

It was predicted that advanced learners would exhibit a frequency effect for both “Chinese-congruent” and “English-only” collocations based on previous empirical research. However, given the low encounter rate of “Chinese-only” collocations across proficiency levels, we expected neither lower-intermediate nor advanced learners to respond significantly to such “Chinese-only” collocations differently under different priming conditions. Previous research on processing L2 collocations by lower-intermediate learners was scarce, making it challenging to predict their sensitivity to corpus frequency.

Statistics revealed that both the lower-intermediate and advanced groups exhibited a response time interaction effect for collocation frequency and type regardless of priming condition (see Figure 3). Post hoc analyses showed distinct patterns:

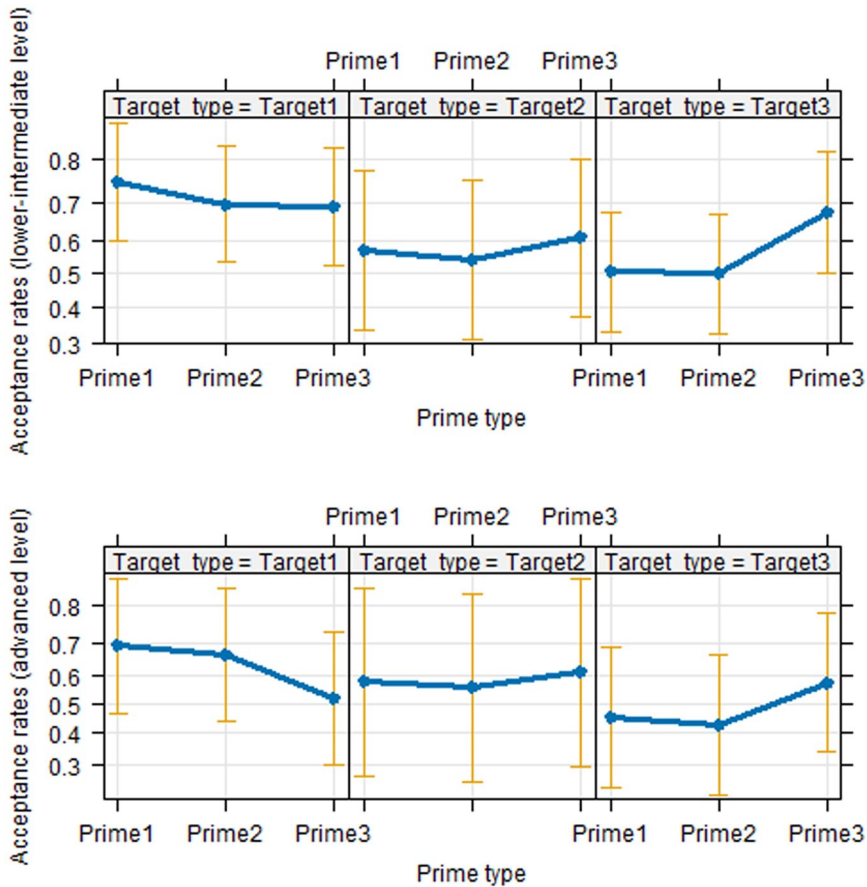


Figure 1. Acceptance rates of the types of English collocations in each priming condition in the lower-intermediate and advanced groups.

Note. Target types (Target1, Target2, and Target3) refer to “L1 Chinese-congruent,” “L1 Chinese-only,” and “L2 English-only” collocations. Prime types (Prime1, Prime2, and Prime3) refer to L1 Chinese-congruent, semantically unrelated, and L2 English-only priming conditions.

Lower-intermediate group: For “Chinese-congruent” collocations, response times decreased as frequency increased. In contrast, there was little change in response times for both “Chinese-only” and “English-only” collocations with increasing frequency. The slope difference between the two types of “Chinese-congruent” and “Chinese-only” collocations was significant ($\beta = -0.22$, $t = -2.6$, $p = .009^{**}$). However, the response time slope differences between “English-only” and “Chinese-only” collocations ($\beta = 0.11$, $t = 1.29$, $p = .196$) and between “English-only” and “Chinese-congruent” collocations ($\beta = -0.11$, $t = -1.26$, $p = .207$) were not significant.

Advanced group: The response time slopes differed significantly between “Chinese-congruent” and “Chinese-only” collocations ($\beta = -0.24$, $t = -2.26$, $p = .024$) and also between “English-only” and “Chinese-only” collocations ($\beta = -0.32$, $t = -2.95$, $p = .003$).

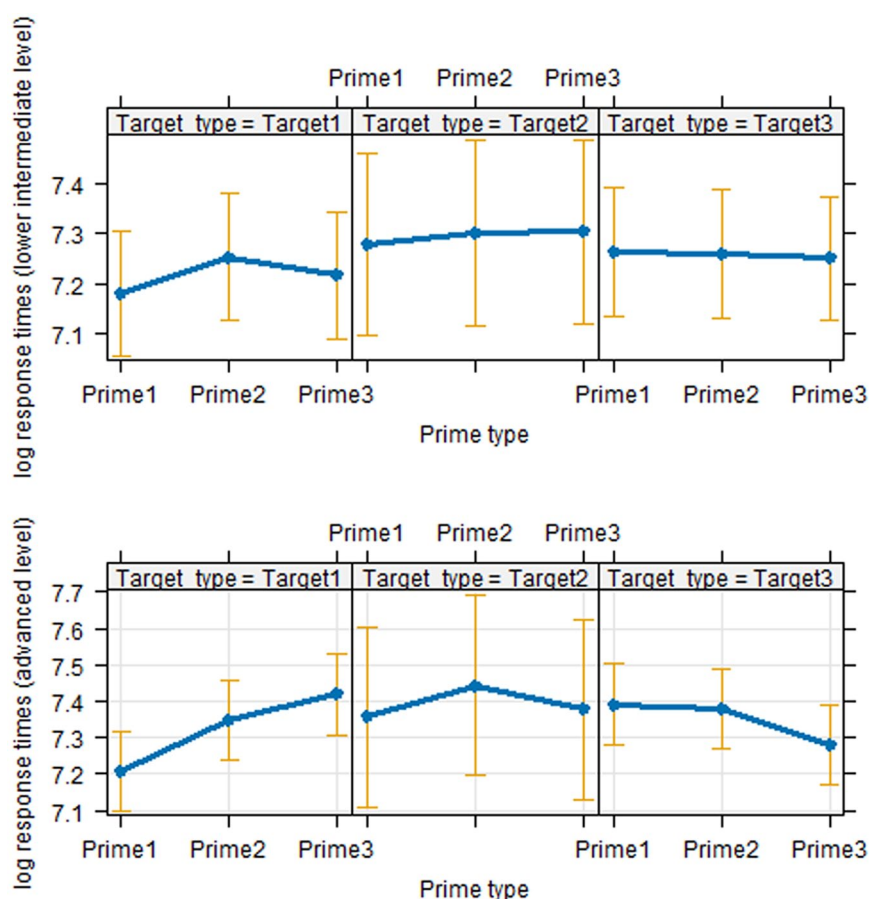


Figure 2. Response times of the types of English collocations in each priming conditions in the lower-intermediate and advanced groups.

Note. Target types (Target1, Target2, and Target3) refer to “L1 Chinese-congruent,” “L1 Chinese-only,” and “L2 English-only” collocations. Prime types (Prime1, Prime2, and Prime3) refer to L1 Chinese-congruent, semantically unrelated, and L2 English-only priming conditions.

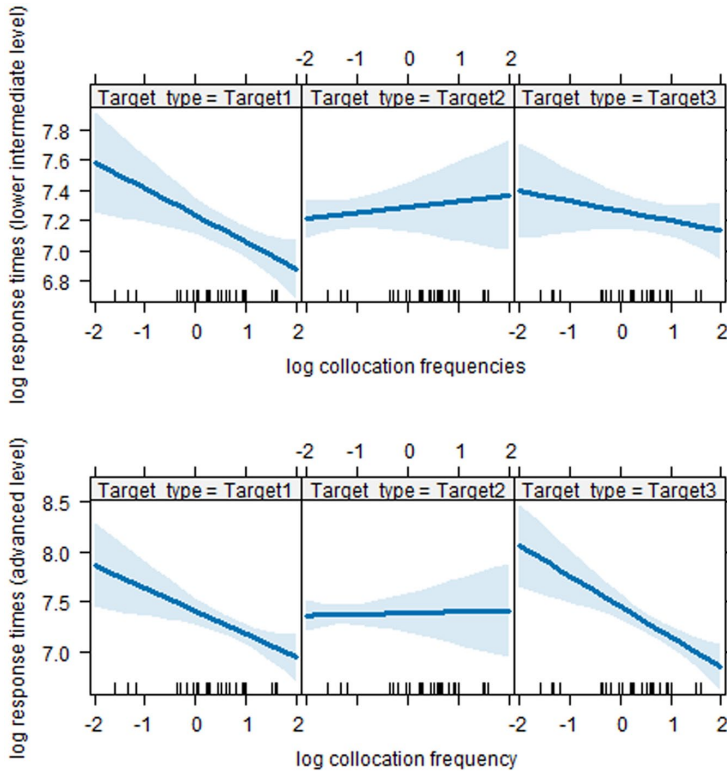
In contrast to the lower-intermediate group, there was no significant difference in response time slopes between “English-only” and “Chinese-congruent” collocations ($\beta = -0.07$, $t = -0.65$, $p = .515$). For both types of collocations, advanced learners showed a decrease in response time with increasing frequency.

These results suggest that the relationship between collocation frequency and response time varies across proficiency levels. Lower-intermediate learners exhibit sensitivity to only “Chinese-congruent” collocation frequency but not for “English-only” or “Chinese-only” ones. In contrast, advanced learners show similar decreases in response time as frequencies of “English-only” as well as “Chinese-congruent” collocations go up.

Table 4. The results of post hoc analyses for unrelated versus translation priming on collocation processing in the respective models by participant group.

Level	Collocation type	Priming conditional contrast	Acceptance rate			Response time		
			Estimate	z	p	Estimate	t	p
Lower-intermediate	"Chinese-only"	Unrelated vs. Chinese congruent	-0.11	-0.59	.557	0.013	0.46	.644
	"Chinese-congruent"	Unrelated vs. Chinese congruent	-0.27	-1.28	.201	0.01	0.46	.644
	"English-only"	Unrelated vs. English-only	-0.74	-3.73	.0002***	-0.001	-0.03	.973
Advanced	"Chinese-only"	Unrelated vs. Chinese congruent	-0.09	-0.37	.713	0.09	1.54	.124
	"Chinese-congruent"	Unrelated vs. Chinese congruent	-0.12	-0.50	.614	0.14	4.00	.0001***
	"English-only"	Unrelated vs. English-only	-0.59	-2.47	.014*	0.10	2.49	.013*

Note. Significant differences are indicated with the convention of * $p < .05$, ** $p < .01$, *** $p < .001$.

**Figure 3.** Response times of the types of English collocations as a function of frequency in the lower-intermediate and advanced groups.

Note. Target types (Target1, Target2, and Target3) refer to "L1 Chinese-congruent," "L1 Chinese-only," and "L2 English-only" collocations.

Discussion

We examined how factors such as L1 Chinese translation priming, the type of L2 English collocation, its frequency, and individuals' L2 proficiency interact to influence the processing of "Chinese-congruent," "Chinese-only," and "English-only" adjective–noun collocations and also investigated whether the different types of English collocations are mentally represented at both lower-intermediate and advanced levels through a cross-language primed acceptability judgment task.

L1 congruency effect, L1 semantic activation, and L2 proficiency

It has been argued that as L2 proficiency increases, the L1 influence on processing—in particular, the L1 congruency effect—weakens. This is supported by studies such as Yamashita and Jiang (2010) and Chen (2023). Our findings support this argument in part: We observed the L1 Chinese congruency effect among advanced learners—specifically, an advantage for "Chinese-congruent" collocations relative to "English-only" ones under the Chinese-congruent priming condition. This was evident in response times between these two types of collocations.

However, contrary to expectations about the initial strong L1 congruency effect for "L1-only" collocations, we found no processing advantage for "L1 Chinese-only" collocations compared to "English-only" ones in both proficiency groups. This result aligns with previous studies (e.g., Lee, 2021; Ren & Wang, 2020; Wolter & Yamashita, 2015, 2018; Zhan & Ni, 2024) and suggests that L1 influence alone does not account fully for the observed congruency effects. If the congruency effect was due to L1 Chinese semantic activation alone, both the "Chinese-congruent" and the almost non-existing "Chinese-only" collocations would have exhibited the same significant congruency effect over the "English-only" collocations even at the lower-intermediate level. The congruency pattern observed challenges the notion that an inherent processing advantage exists purely because of L1 semantic knowledge, as was posited in L1 semantic activation theory (Jiang, 2000a).

It is true that learners at an early stage are particularly susceptible to L1 influence (Sunderman & Kroll, 2006). However, our results indicate that this susceptibility does not always translate into a processing advantage. Other factors—such as actual L2 exposure and proficiency—are likely more influential. This perspective is consistent with Sonbul & El-Dakhs (2020), Wolter and Gyllstad (2013), and Yamashita and Jiang (2010), who argue that L1 congruence effects are not solely due to L1 semantic activation but also depend on actual exposure to L2 collocations.

L2 collocation knowledge representation and proficiency

According to the L1 semantic activation theory, lower-level L2 learners are expected to have stronger mental representations for "L1-congruent" and "L1-only" collocations compared to "L2-only" ones. However, our analyses with the lower-intermediate group revealed a different picture: Neither "Chinese-congruent," nor "Chinese-only," nor "English-only" collocations showed significant differences in processing time across all three priming conditions (Chinese-congruent, English-congruent, and semantically unrelated). This suggests that there is an overall lack of robust mental representations for these types of collocations among lower-intermediate learners.

In contrast, with the advanced group, we observed a notable difference: "Chinese-congruent" and "English-only" collocations showed shorter response times when presented in their respective translation priming conditions compared to the semantically unrelated condition. This indicates that both types of collocations have significantly enhanced representations at this proficiency level.

These findings suggest that L2 collocation representations are not necessarily established automatically through activating L1 semantics alone at the start but rather develop gradually during L2

learning. This perspective aligns with previous research on formulaic expressions, as highlighted by Tremblay et al. (2011) and Cao (2016, 2018).

Tremblay et al. (2011) tested out that the processing advantage of formulaic expressions is attributable to their holistic representation as a single lexical unit. According to Cao's (2016, 2018) studies, this holistic representational property is not intrinsic but acquired gradually over time. Cao found no processing advantage of formulaic over non-formulaic expressions by low-level L2 English learners, whereas advanced learners exhibited such an advantage. This indicates the incremental development of mental representations in formulaic expressions.

Thus, it appears that L2 collocation knowledge is not automatically derived from L1 semantics but rather develops gradually as part of a broader process of L2 learning and acquisition.

L2 collocation frequency, type, and proficiency

This study also demonstrated an interaction effect between collocation type and frequency among both lower-intermediate and advanced learners, regardless of priming conditions. Lower-intermediate learners were significantly sensitive to the frequencies of "Chinese-congruent" collocations but not to those of "Chinese-only" and "English-only" ones, while advanced learners showed a similar sensitivity pattern for both "Chinese-congruent" and "English-only" collocations.

These findings align with previous research on university L2 English learners (Wolter & Gyllstad, 2013; Wolter & Yamashita, 2018; Zhang & Fang, 2020), which observed frequency effects for both "L1-congruent" and "L2-only" collocations. However, it is important to note that the term "frequency" refers to native-speaker corpus frequency, not learner exposure frequency.

Wolter and Gyllstad (2013) emphasized that while lexical frequency in a native-speaker corpus reflects general mental lexicons of majority speakers, individual learners' frequency sensitivity may differ. Learners require sufficient exposure before becoming sensitive to certain collocation types. Thus, the degree of sensitivity indirectly indicates their level of L2 experience with specific collocations beyond an exposure threshold.

The observed sensitivity pattern supports the notion that L1 congruency effects are primarily influenced by frequency of exposure over time in L2 learning. Lower-intermediate learners may have reached the exposure threshold for "Chinese-congruent" collocations only, which explains their sensitivity to "Chinese-congruent" frequencies but not "English-only" ones.

The necessity for actual L2 exposure is crucial for establishing a link with L1 semantics; only then can an equivalent literal-translation or free-translation L1 prime activate L2 collocation processing, as evidenced in our study. This aligns with Wolter and Gyllstad's (2013) findings, which suggest that lower-level learners may not have reached the exposure threshold of corpus frequency sensitivity for "L2-only" collocations as they did for L1-congruent ones.

Du et al.'s (2023) L2 multiword expression experience account further emphasizes this point by noting that prior exposure to a word sequence in an L2 is essential for faster processing, regardless of whether it has a word-for-word/literal equivalent form in the L1. However, having a word-for-word L1 translation equivalent may facilitate learning of L2 multiword expressions; acquiring L1 congruent multiword expressions typically takes less time and requires fewer exposures compared to incongruent L2-only ones, which contributes to the congruency effect.

In summary, our results and previous findings underscore the importance of experience with specific L2 collocation types over time rather than L1 influence on processing advantages. This aligns with Kroll and Bialystok's (2013) research highlighting robust cross-language activations in proficient bilinguals who have gained extensive exposure to their L2.

Conclusion

Using a cross-language primed collocation acceptability judgment paradigm, this study examined processing effects of L2 English collocations in different L1 Chinese semantic priming conditions with lower-intermediate and advanced Chinese-English bilinguals. The results revealed several key findings that complement the traditional view on L1 congruency effects:

First, contrastive processing effects between “Chinese-congruent” versus “English-only” collocations and “Chinese-only” versus “English-only” collocations at the lower-intermediate and advanced levels suggest that L1 influence is not solely responsible for the congruency effect. Factors beyond mere semantic congruency may also be involved.

Second, lower-intermediate learners exhibited an overall lack of mental representation for all types of collocations compared to advanced learners, who showed enhanced representations for encountered “Chinese-congruent” and “English-only” collocations but persistent lack of representation for the almost non-existent “L1 Chinese-only” collocations. This finding underscores that L2 collocation representations do not simply arise from the availability of equivalent L1 semantics; instead, they are gradually established through consistent exposure to specific types of collocations over time.

Third, response times decreased as a function of higher collocation frequency for “L1 Chinese-congruent” collocations across both proficiency groups, indicating a general frequency effect in processing L2 collocations. However, the frequency sensitivity for “L2 English-only” collocations did not appear until the advanced level, which suggests that actual exposure to specific types of L2 collocations is crucial for the frequency effect to become visible.

In conclusion, our results provide evidence supporting the idea that the processing and mental representation of L2 collocations depends on essential L2 exposure rather than immediate L1 semantic activation. This has significant implications for foreign language teaching: formal instruction should focus on providing ample opportunities for learners to encounter specific collocations in context.

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Ethical consideration

The School of Foreign Studies Ethics Review Committee at South China Normal University approved our language lab experiments on March 10, 2021.

Consent to Participate

Participants gave written consent for review and signature before starting language experiments.

Author contributions

Jianyao Yu: Project administration; Conceptualization; Methodology; Visualization; Data curation; Formal analysis; Writing—original draft; Writing—review and editing.

Ping Zhang: Methodology; Second draft revision; Funding acquisition; Supervision; Writing—review and editing.

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The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

Data will be made available on request.

Supplemental material

Supplemental material for this article is available online.

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